COMPUTER SYSTEMS AND PLATFORM TECHNOLOGIES

IT0047

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Technology Driven by Innovation

MODULE 1

Data Manipulation

MODULE 1.1

Number System - Conversion

At the end of the lesson, the student should be able to :

- Identify the different number system as they relate to computers
- Identify the unit, number, base/radix, positional notation, least significant digit and most significant digit
- Convert values from any number system to each other and back to other systems

Number Base

- What is a **number base**?

  A **number base** is a specific collection of symbols on which a number system can be built.

- The number base familiar to us is *base 10*, upon which the *decimal number system* is built. There are ten symbols - 0 to 9 - used in the decimal system.

Place Value

- What is the concept of **place value**?

  Place value means that the value of a digit in a number depends not only on its own natural value but also on its location in the number.

  It is used interchangeably with the term **positional notation**.

- Place value tells us that the two 4s in the number **3474** have different values, that is, **400**

  and **4**, respectively.

A Review of the Decimal Number System

- The word “decimal” comes from the Latin word *decem*, meaning *ten*.
- Thus, the number base of the decimal number system is *base 10*.
- Since it is in base 10, *ten symbols* are used in the decimal number system.

  {0,1,2,3,4,5,6,7,8,9}
- This means that **only** the digits in the above set can be used for each position in every place

  value in a given decimal number.

A Review of the Decimal Number System

2 7 **0**

- Note that the highlighted place value can be filled by the digits in the set {0,1,2,3,4,5,6,7,8,9}.
- Thus, it can be increased by 1 until it reaches -

2 7 **9**

- At this point, the symbols that can be used to fill the highlighted position has been exhausted. Increasing it further causes a shift in place value and resets the initial place value to zero. Thus -

2 **8** 0

A Review of the Decimal Number System

Case Study: **3474**

Using place values, the number 3474 is understood to mean,

$$3000 + 400 + 70 + 4 = 3474$$

This can also be expressed as –

$$(3x1000) + (4x100) + (7x10) + 4 = 3474$$

Note that each digit is multiplied by **powers of 10**, so that the above is equal to –

$$\mathbf{(3x10^3) + (4x10^2) + (7x10^1) + (4x10^0) = 3474}$$

Note that the rightmost exponent starts from zero and increases by 1 as the place value increases.

Hence, the decimal number system is said to be in ***base 10****.*

Base 2: The Binary Number System

- The word “binary” comes from the Latin word *bis*, meaning *double*.
- Thus, the number base of the binary number system is *base 2*.
- Since it is in base 2, *two symbols* are used in the binary number system.

  {0,1}
- This means that **only** the digits in the above set can be used for each position in every place value in a given binary number.

Base 2: The Binary Number System

1	1	0	**0**

- Note that the highlighted place value can be filled by the digits in the set {0,1}.
- Thus, it can be increased by 1 until it reaches -

1	1	0	**1**

- At this point, the symbols that can be used to fill the highlighted position has been exhausted. Increasing it further causes a shift in place value, and resets the initial place value to zero. Thus -

1	1	**1**	0

Base 2: The Binary Number System

- To avoid confusion, one should write a binary number with *base 2* as its subscript whenever necessary.
- Thus, the binary number **10110** should be written as -

$$\mathbf{10110_2}$$

- It should be read as *"one-zero-one-one-zero base two"* and **NOT** *"ten-thousand one-hundred ten"* since each phrase denotes an entirely different number.

Base 2: The Binary Number System

Case Study: $\mathbf{10110_2}$

- We know that the decimal number 3474 can be expressed as powers of 10 –

$$(3x10^3) + (4x10^2) + (7x10^1) + (4x10^0) = 3474_{10}$$

- In the same manner, the binary number $\mathbf{10110_2}$ can be expressed as **powers of 2** –

$$\mathbf{(1x2^4) + (0x2^3) + (1x2^2) + (1x2^1) + (0x2^0) = 22_{10}}$$

Note that the rightmost exponent starts from zero and increases by 1 as the place value increases.

Hence, the binary number system is said to be in ***base 2****.*

Base 8: The Octal Number System

- The word “octal” comes from the Greek word *oktõ*, meaning *eight*.
- Thus, the number base of the octal number system is *base 8*.
- Since it is in base 8, *eight symbols* are used in the octal number system.

  {0,1,2,3,4,5,6,7}
- This means that **only** the digits in the above set can be used for each position in every place

  value in a given octal number.

Base 8: The Octal Number System

3	6	**0**

- Note that the highlighted place value can be filled by the digits in the set {0,1,2,3,4,5,6,7}.
- Thus, it can be increased by 1 until it reaches -

3	6	**7**

- At this point, the symbols that can be used to fill the highlighted position has been exhausted. Increasing it further causes a shift in place value, and resets the initial place value to zero. Thus -

3	**7**	0

Base 8: The Octal Number System

- To avoid confusion, one should write an octal number with *base 8* as its subscript whenever necessary.
- Thus, the octal number **72143** should be written as -

$$\mathbf{72143_8}$$

- It should be read as *"seven-two-one-four-three base eight"* and **NOT** *"seventy two-thousand one-hundred forty three"* since each phrase denotes an entirely different number.

Base 8: The Octal Number System

Case Study: $\mathbf{72143_8}$

- We know that the decimal number 3474 can be expressed as powers of 10 –

$$(3x10^3) + (4x10^2) + (7x10^1) + (4x10^0) = 3474_{10}$$

- In the same manner, the octal number $\mathbf{72143_8}$ can be expressed as **powers of 8** –

$$\mathbf{(7x8^4) + (2x8^3) + (1x8^2) + (4x8^1) + (3x8^0) = 29795_{10}}$$

Note that the rightmost exponent starts from zero and increases by 1 as the place value increases.

*Hence, the octal number system is said to be in **base 8**.*

Base 16: The Hexadecimal Number System

- The word “hexadecimal” is a combination of the Greek word *hex*, meaning *six* and the Latin word *decem*, meaning *ten*.
- Thus, the number base of the hexadecimal number system is *base 16*.
- Since it is in base 16, *sixteen symbols* are used in the hexadecimal number system.

  {0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}
- This means that **only** the digits in the above set can be used for each position in every place value in a given hexadecimal number.

Base 16: The Hexadecimal Number System

A 3 B **0**

- Note that the highlighted place value can be filled by the digits in the set {0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}.
- Thus, it can be increased by 1 until it reaches -

A 3 B **F**

- At this point, the symbols that can be used to fill the highlighted position has been exhausted. Increasing it further causes a shift in place value, and resets the initial place value to zero. Thus -

A 3 **C** 0

Base 16: The Hexadecimal Number System

- To avoid confusion, one should write a hexadecimal number with *base 16* as its subscript whenever necessary.
- Thus, the hexadecimal number **B23C** should be written as -

$$\mathbf{B23C_{16}}$$

- It should be read as *"b-two-three-c base sixteen".*

Base 16: The Hexadecimal Number System

Case Study: **$B23C_{16}$**

- We know that the decimal number 3474 can be expressed as powers of 10 –

$$(3x10^3) + (4x10^2) + (7x10^1) + (4x10^0) = 3474_{10}$$

- In the same manner, the hexadecimal number **$B23C_{16}$** can be expressed as **powers of 16** –

$$\mathbf{(11x16^3) + (2x16^2) + (3x16^1) + (12x16^0) = 45628_{10}}$$

Note that the rightmost exponent starts from zero and increases by 1 as the place value increases.

*Hence, the hexadecimal number system is said to be in **base 16**.*

Comparative Values: Bases 10, 2, 8, 16

Base 10	Base 2	Base 8	Base 16
Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8

Comparative Values: Bases 10, 2, 8, 16

Base 10	Base 2	Base 8	Base 16
Decimal	Binary	Octal	Hexadecimal
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F
16	10000	20	10
17	10001	21	11

Conversion from Bases 2, 8, 16 to Base 10

- <u>Technique</u>: *Expand the number using the appropriate base.*
- Convert $\mathbf{11010_2}$ to base 10.

$$(1x2^4) + (1x2^3) + (0x2^2) + (1x2^1) + (0x2^0) = ?$$

$$(1x16) + (1x8) + (0x4) + (1x2) + (0x1) = ?$$

$$16 + 8 + 0 + 2 + 0 = \mathbf{26_{10}}$$

- Convert $\mathbf{1537_8}$ to base 10.

$$(1x8^3) + (5x8^2) + (3x8^1) + (7x8^0) = ?$$

$$(1x512) + (5x64) + (3x8) + (7x1) = ?$$

$$(512) + (320) + (24) + (7) = \mathbf{863_{10}}$$

Conversion from Bases 2, 8, 16 to Base 10

- Technique: *Expand the number using the appropriate base.*

- Convert **$AF4_{16}$** to base 10.

$$(A \times 16^2) + (F \times 16^1) + (4 \times 16^0) = ?$$

$$(10 \times 256) + (15 \times 16) + (4 \times 1) = ?$$

$$2560 + 240 + 4 = \mathbf{2804_{10}}$$

Conversion from Base 10 to Bases 2, 8, 16

- Technique: *Divide the number using the appropriate base, then repeatedly divide the resulting quotients by the same base until it becomes zero. The answer is the combined remainders in reverse order accumulated from the divisions. This technique is called the* ***remainder method****.*
- Convert $\mathbf{133_{10}}$ to base 2.

133 ÷ 2 = 66	r. 1
66 ÷ 2 = 33	r. 0
33 ÷ 2 = 16	r. 1
16 ÷ 2 = 8	r. 0
8 ÷ 2 = 4	r. 0
4 ÷ 2 = 2	r. 0
2 ÷ 2 = 1	r. 0
1 ÷ 2 = **0**	r. 1

Answer: $\mathbf{10000101_2}$

Conversion from Base 10 to Bases 2, 8, 16

- Technique: *Divide the number using the appropriate base, then repeatedly divide the resulting quotients by the same base until it becomes zero. The answer is the combined remainders in reverse order accumulated from the divisions. This technique is called the* ***remainder method****.*
- Convert $\mathbf{7863_{10}}$ to base 8.

7863 ÷ 8 = 982	r. 7
982 ÷ 8 = 122	r. 6
122 ÷ 8 = 15	r. 2
15 ÷ 8 = 1	r. 7
1 ÷ 8 = **0**	r. 1

Answer: $\mathbf{17267_8}$

Conversion from Base 10 to Bases 2, 8, 16

- Technique: *Divide the number using the appropriate base, then repeatedly divide the resulting quotients by the same base until it becomes zero. The answer is the combined remainders in reverse order accumulated from the divisions. This technique is called the* ***remainder method****.*

- Convert $\mathbf{2926_{10}}$ to base 16.

  2926 ÷ 16 = 182 r. 14(E)

  182 ÷ 16 = 11 r. 6

  11 ÷ 16 = **0** r. 11(B)

Answer: $\mathbf{B6E_{16}}$

Conversion from Base 8 to Base 2

- Technique: *Convert each octal digit to a set of three binary digits.*

- Convert $\mathbf{5641_8}$ to base 2.

5	**6**	**4**	**1**
101	110	100	001

Answer: $\mathbf{101110100001_2}$

Conversion from Base 2 to Base 8

- <u>Technique</u>: *Convert each set of three binary digits to an octal digit starting from the right going to the left, adding leading zeroes if necessary.*
- Convert $\mathbf{1111010_2}$ to base 8.

001	**111**	**010**
1	7	2

Answer: $\mathbf{172_8}$

Conversion from Base 16 to Base 2

- Technique: *Convert each hexadecimal digit to a set of four binary digits.*
- Convert **$A3CF_{16}$** to base 2.

A	**3**	**C**	**F**
1010	0011	1100	1111

Answer: **1010001111001111_2**

Conversion from Base 2 to Base 16

- Technique: *Convert each set of four binary digits to a hexadecimal digit starting from the right going to the left, adding leading zeroes if necessary.*

- Convert **101011010_2** to base 16.

0001 0101 1010

1 5 A

Answer: **$15A_{16}$**

MODULE 1.2

Number System – Arithmetic Operation

At the end of the lesson, the student should be able to:

- Perform Addition, Subtraction, Multiplication and Division of binary, octal and hexadecimal

Decimal Number (Base 10)

- Similar to the ordinary counting numbers from 0 to 9.
- Arithmetic operations such as: Addition, Subtraction, Multiplication and Division are done of the same manner/procedure with performing mathematical operations on counting numbers.

Binary Number (Base 2)

- Binary numbers represents only two numbers , (0 and 1).
- There are different rules in performing arithmetic operations on binary numbers.0

Binary Number (Base 2)

Binary Addition:

0 + 0 = 0
0 + 1 = 1
1 + 0 = 1
1 + 1 = 0 carry 1

Example:

```
   1011101
+   1101100
  --------
  11001001
```

Binary Number (Base 2)

Binary Subtraction:

0 – 0 = 0
0 – 1 = 1 borrow 1
1 – 0 = 1
1 – 1 = 0

Example:

```
   1001101
-  1101000
   -------
   1100101
```

Binary Number (Base 2)

Binary Multiplication:

0 * 0 = 0
0 * 1 = 0
1 * 0 = 0
1 * 1 = 0

Example:

```
     1001101
*        110
  0000000
    1001101
   1001101
  111001110
```

Binary Number (Base 2)

Binary Division:

- Check the number of digits of the divisor; eg. 3, then take the first 3 digits of the dividend.
- f the dividend is greater than the divisor, then the answer is 1, otherwise the answer is 0.
- Proceed with the process of subtraction, bringing down the next bit of the dividend. Remember you are subtracting binary numbers. Then repeat the process.

Binary Number (Base 2)

Binary Division:

Example:

1001101 / 110

100 / 110 = 0 , 100 – 000 = 100

1001 / 110 = 1, 1001 – 110 = 0011, bring down 1 00111 / 110 = 1,

111 – 110 = 001, bring down 0

0010 / 110 = 0, 0010 – 000 = 10, bring down 1

101 / 110 = 0, 101 – 000 = 101

answer: 1100 r. 101

Octal Number (Base 8)

- Octal numbers represents numbers from (0 to 7).
- There are different rules in performing arithmetic operations on octal numbers.

Octal Number (Base 8)

Octal Addition:

In adding octal numbers, if the result is greater than 7, it should be subtracted by 8 until the result is less than 8. The number of times you subtracted 8, will be the number of carry on the next bit.

Octal Number (Base 8)

Octal Addition:

Example:

a)

$$\begin{array}{r} 2345 \\ +\ 1736 \\ \hline \mathbf{4303} \end{array}$$

b)

$$\begin{array}{r} 7632 \\ +\ 5774 \\ \hline \mathbf{15626} \end{array}$$

Octal Number (Base 8)

Octal Subtraction:

In subtracting octal numbers, if the number cannot be subtracted then it needs to borrow bit from the next bit, every bit borrowed will be equivalent to 8 and will be added to the bit that needs it, then the process of subtraction will be done.

Octal Number (Base 8)

Octal Subtraction:

Example:

a)

$$\begin{array}{r} 2345 \\ -\ 1736 \\ \hline \mathbf{0407} \end{array}$$

b)

$$\begin{array}{r} 7632 \\ -\ 5774 \\ \hline \mathbf{1636} \end{array}$$

Octal Number (Base 8)

Octal Multiplication:

In multiplying octal numbers, it also follows the procedure in multiplying decimal numbers, every bit in the multiplier will be multiplied to the multiplicand, if the answer is greater than 7, then subtract 8, until the answer is less than 8. The number of times you subtracted 8 is the number to be carried to the next bit. Answers are arranged just like in multiplying decimal numbers, to obtain the final answer, proceed with the addition process, however remember that you are adding octal numbers.

Octal Number (Base 8)

Octal Multiplication:

Example:

a)

```
     2345
*      36
    16536
  + 7257
   111326
```

b)

```
     7632
*      74
    37150
  + 66266
   722030
```

Octal Number (Base 8)

Octal Division:

- In dividing octal numbers, it also follows the procedure in dividing decimal numbers, Check the number of digits of the divisor; eg. 3, then take the first 3 digits of the dividend.
- Proceed with the process of subtraction, bringing down the next bit of the dividend. Then repeat the process. Remember that you are subtracting octal numbers.

Octal Number (Base 8)

Octal Division:

Example:

27 / 7 = 3, 7 * 3 in octal is 25
27 – 25 = 2, 2 cannot be divided by 7

Answer is: 3 remainder 2

Hexadecimal Number (Base 16)

- Hexadecimal numbers represents numbers from (0 to 15).
- However, numbers from 10 to 15 are represent by letters for A to F respectively.

- There different rules in performing arithmetic operations on hexadecimal numbers.

Hexadecimal Number (Base 16)

Hexadecimal Addition:

In adding hexadecimal numbers, if the result is greater than 15, it should be subtracted by 16 until the result is less than 16. The number of times you subtracted 16, will be the number of carry on the next bit.

Hexadecimal Number (Base 16)

Hexadecimal Addition:

Example:

a)

```
   B34A
+  1736
  ----
   CA80
```

b)

```
   F6E2
+  5974
  -----
  15056
```

Hexadecimal Number (Base 16)

Hexadecimal Subtraction:

In subtracting hexadecimal numbers, if the number cannot be subtracted then it needs to borrow bit from the next bit, every bit borrowed will be equivalent to 16 and will be added to the bit that needs it, then the process of subtraction will be done.

Hexadecimal Number (Base 16)

Hexadecimal Subtraction:

Example:

a)

```
  2A45
- F73B
  330A
```

b)

```
  E632
- 5C7A
  89B8
```

Hexadecimal Number (Base 16)

Hexadecimal Multiplication:

In multiplying hexadecimal numbers, it also follows the procedure in multiplying decimal numbers, every bit in the multiplier will be multiplied to the multiplicand, if the answer is greater than 15, then subtract 16, until the answer is less than 16. The number of times you subtracted 16 is the number to be carried to the next bit. Answers are arranged just like in multiplying decimal numbers, to obtain the final answer, proceed with the addition process, however remember that you are adding hexadecimal numbers.

Hexadecimal Number (Base 16)

Hexadecimal Multiplication:

Example:

a)

```
    B34A
*     36
  433BC
+219DE
 25D19C
```

b)

```
    7632
*     74
   D8C8
+33B5E
  348EA8
```

Hexadecimal Number (Base 16)

Hexadecimal Division:

- In dividing hexadecimal numbers, it also follows the procedure in dividing decimal numbers, Check the number of digits of the divisor; eg. 3, then take the first 3 digits of the dividend.
- Proceed with the process of subtraction, bringing down the next bit of the dividend. Then repeat the process. Remember that you are subtracting hexadecimal numbers.

Hexadecimal Number (Base 16)

Hexadecimal Division:

Example:

27 / 7 = 5, 7 * 5 in hexadecimal is 23
27 – 23 = 4, 4 cannot be divided by 7

Answer is: 5 remainder 4

MODULE 1. 3

Number System – Digital Codes & Parity

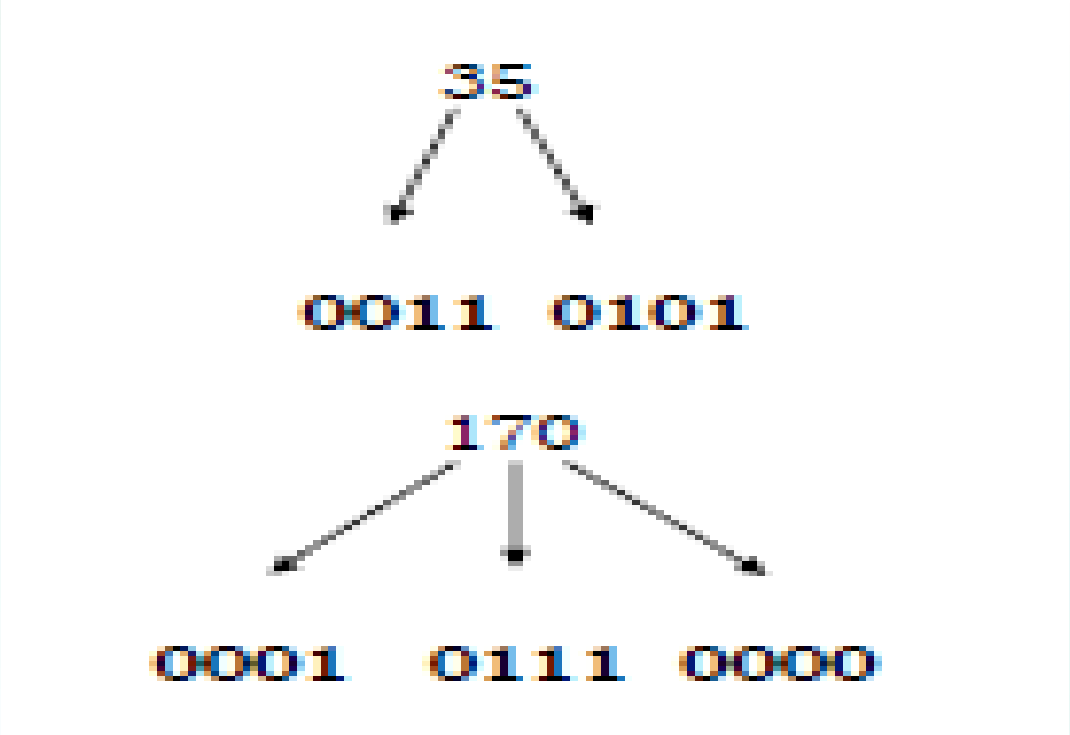
At the end of the lesson, the student should be able to :

- **convert decimal numbers into digital codes (8421, 84-2-1, 2421, Excess-3, Biquinary, and Gray codes)**
- **identify decimal numbers represented by the digital codes**
- **distinguish Even from Odd parity**

The 8421 Code

- It is a type of **binary coded decimal (BCD) code. Binary coded decimal means that each decimal digit, 0 to 9, is represented by the binary code of four bits.**
- The designation 8421 indicates the binary weights of the four bits (23, 22, 21, 20).
- To express any decimal number in BCD, simply replace each decimal digit with the appropriate 4-bit code.

The 8421 Code



The 8421 Code

You should realize that, with four bits, sixteen numbers (0000 through 1111) can be represented but that, in the 8421 code, only ten of these are used. The six code combinations that are not used – 1010, 1011, 1100, 1101, 1110, and 1111 – are **invalid in 8421 code.**

The Excess-3 Code

This is unweighted code; its code assignment is obtained from the corresponding value of BCD after the addition of three.

The 84-2-1 and 2421 Codes

These are weighted code. The designation 8421 and 2421 indicates the binary weights of the four bits.

The Biquinary Code

- It is an example of a seven-bit code with error-detection properties. Each decimal digit consists of five 0's and two 1's placed in the corresponding weighted columns (5043210).
- During transmission of signals from one location to another, an error may occur. One or more bits may change value. A circuit in the receiving side can detect the presence of more (or less) than two 1's and if the received combination of bits does not agree with the allowed combination, an error is detected.

Digital Codes

Decimal	BCD	Excess 3	8 4 -2 -1	2 4 2 1	Biquinary
0	0000	0011	0000	0000	01 00001
1	0001	0100	0111	0001	01 00010
2	0010	0101	0110	0010	01 00100
3	0011	0110	0101	0011	01 01000
4	0100	0111	0100	0100	01 10000
5	0101	1000	1011	1011	10 00001
6	0110	1001	1010	1100	10 00010
7	0111	1010	1001	1101	10 00100
8	1000	1011	1000	1110	10 01000
9	1001	1100	1111	1111	10 10000

The Gray Code

- It is unweighted and is not arithmetic code: that is, there are no specific weights assigned to the bit positions.

- The important feature of the Fray Code is that *it exhibits only a single bit change from one code to the next.*

Binary-to-Gray Code Conversion

1. The most significant bit (left-most) in the Gray code is the same as the corresponding MSB in the binary number.
2. Going from left to right, add each adjacent pair of binary code bits to get the next Gray code bit. Discard carries.

Binary-to-Gray Code Conversion

Example:

1	+ →	**0**	+ →	**1**	+ →	**1**	+ →	**0**	Binary
↓		↓		↓		↓		↓	
1		**1**		**1**		**0**		**1**	Gray code

Decimal	Binary	Gray Code	Decimal	Binary	Gray Code
0	0000	0000	8	1000	1100
1	0001	0001	9	1001	1101
2	0010	0011	10	1010	1111
3	0011	0010	11	1011	1110
4	0100	0110	12	1100	1010
5	0101	0111	13	1101	1011
6	0110	0101	14	1110	1001
7	0111	0100	15	1111	1000

Gray Code-to-Binary Conversion

1. The most significant bit (left-most) in the Gray code is the same as the corresponding MSB in the binary number.
2. Add each binary code bit generated to the gray code bit in the next adjacent position. Discard carries.

Gray Code-to-Binary Conversion

Example:

1		1		0		1		0	Gray code
↓	+ ↗	↓	+ ↗	↓	+ ↗	↓	+ ↗	↓	
1		0		0		1		0	Binary

Parity

- It is one of the most common ways to achieve error detection.
- A *parity bit is an extra bit included with a message to make the total number of 1's transmitted either odd or even.*
- ***Odd Parity: The parity bit is chosen so that the total number of 1s is odd. Even Parity: The parity bit is chosen so that the total number of 1s is even.***
- It can detect odd number of errors but not even number of errors.

Parity

Examples:

	Odd Parity	Even Parity
0000	1	0
0001	0	1
0010	0	1
0011	1	0
0100	0	1
0101	1	0

7 ASCII bits + 1 Parity bit = 8 bits = 1 byte.

A p100 0001
(p = 0 for even parity.)| (p = 1 for odd parity.)

B p100 0010
(p = 0 for even parity.) (p = 1 for odd parity.)

C p100 0011
(p = 1 for even parity.) (p = 0 for odd parity.)

Parity

Error Detection Example

Send:
A (0100 0001) with even parity.

Receive:
Should receive (0100 0001) an even number of 1s.

If an odd number of 1s is received, there is an error.
0101 0001 - 1 error: Can be detected with parity.
0111 0001 - 2 errors: Cannot detect with parity.
0111 0011 - 3 errors: Can be detected with parity.
0111 0010 - 4 errors: Cannot detect with parity.
0101 1110 - 5 errors: Can be detected with parity.

Binary, Signed Integer Representations

Three systems are widely used for representing both positive and negative numbers:

- 1. Sign and Magnitude
- 2. 1's – complement
- 3. 2's – complement

- For all three systems, the leftmost bit is 0 for positive numbers and 1 is for negative numbers

Binary, Signed Integer Representations

$b_3\ b_2\ b_1\ b_0$	Sign and Magnitude	1's Complement	2's Complement
0000	+ 0	+ 0	+ 0
0001	+ 1	+ 1	+ 1
0010	+ 2	+ 2	+ 2
0011	+ 3	+ 3	+ 3
0100	+ 4	+ 4	+ 4
0101	+ 5	+ 5	+ 5
0110	+ 6	+ 6	+ 6
0111	+ 7	+ 7	+ 7
1000	- 0	- 7	- 8
1001	- 1	- 6	- 7
1010	- 2	- 5	- 6
1011	- 3	- 4	- 5
1100	- 4	- 3	- 4
1101	- 5	- 2	- 3
1110	- 6	- 1	- 2
1111	- 7	- 0	- 1

Be reminded that:

- Positive values have identical representations in all systems.
- In the sign-and-magnitude system, negative values are represented by changing the MSB of the positive representation to 1.

Be reminded that:

- In the 1’s-complement system, negative values are obtained by complementing each bit of the corresponding positive representation.

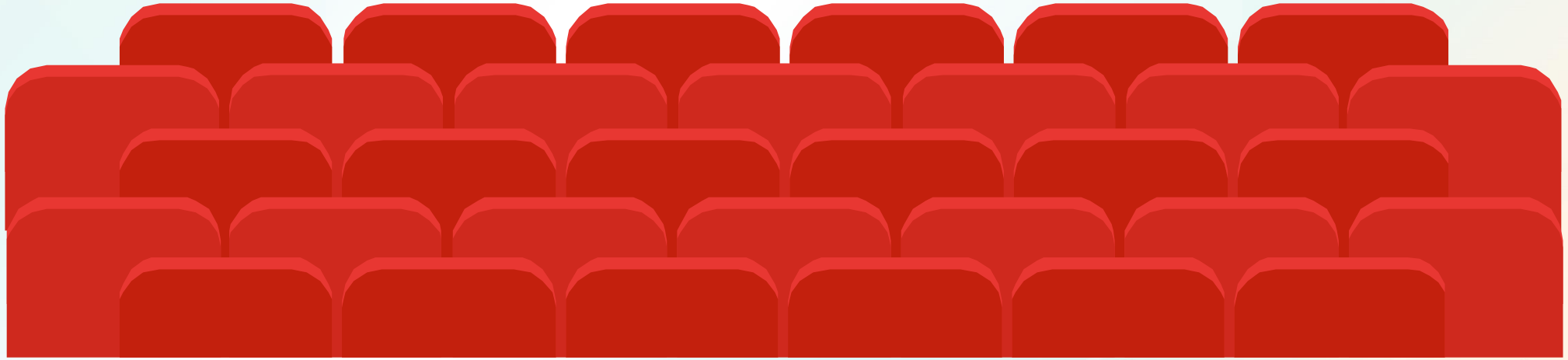
- In the 2’s-complement system, a negative number is obtained by subtracting the positive number from 2n. Hence, the 2’s-complement representation is obtained by adding 1 to the 1’s-complement representation.

Overflow and Underflow

- Overflow (Addition)
  - MSB - Carry, No Carry
  - MSB – No Carry, Carry

- Underflow (Subtraction)
  - MSB – Borrow, No Borrow
  - MSB – No Borrow, Borrow

Q&A SESSION

ASK ANY QUESTION RELATED TO OUR TOPIC FOR TODAY.

THANK YOU